Check for updates

Wiley
International Journal of Energy Research
Volume 2026, Article ID 1140262, 23 pages
https://doi.org/10.1155/er/1140262

WILEY

Review Article

Forecasting the Potential, Power and Energy (PPE) of Both Building Integrated PV and Traditional PV (BIPV–PV) Systems Using State-of-the-Art AI Methods

Mohammed Sadeq, Firdaus Muhammad-Sukki, and Nazmi Sellami

School of Computing, Engineering and the Built Environment, Edinburgh Napier University, Merchiston Campus, Edinburgh, UK

Correspondence should be addressed to Mohammed Sadeq; mohammed.abdulhameed@napier.ac.uk and Nazmi Sellami; n.sellami@napier.ac.uk

Received 29 July 2025; Revised 1 December 2025; Accepted 20 December 2025

Academic Editor: Da Xu

Copyright © 2026 Mohammed Sadeq et al. International Journal of Energy Research published by John Wiley & Sons Ltd. This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

Forecasting the potential and output of building-integrated photovoltaic (BIPV) and traditional photovoltaic (PV) systems, including rooftop, ground-mounted and industrial-shed installations, has become increasingly important, as these technologies hold substantial potential for meeting a significant share of energy demand. Artificial intelligence (AI) approaches, including machine learning (ML) and deep learning (DL) models, are widely recognised as powerful tools for forecasting solar resource potential and system performance. These models play an essential role in accelerating the integration of renewable energy within urban energy planning frameworks. In this context, forecasting for BIPV–PV systems can be broadly classified into three domains: potential, power and energy (PPE). Given the rapid advances in the field of DL over the past few years, numerous studies have made targeted efforts to improve the forecasting accuracy for both BIPV–PV systems by enhancing input data quality and applying advanced, complex and hybrid models. Most of these efforts have mainly narrowed their focus to one of the three forecasting domains rather than adopting a more integrated approach. This systematic literature review (SLR) aims to provide a comprehensive review of PPE forecasting approaches to enable more robust assessment and deeper insights into the feasibility and viability of BIPV–PV systems. The review further highlights key methodological challenges, outlines limitations and offers practical guidance for researchers, policymakers and developers, while identifying emerging trends and future opportunities in AI-based forecasting for BIPV–PV applications.

Keywords: artificial intelligence (AI); building-integrated PV (BIPV); deep learning (DL); grid-connected PV systems; hybrid forecasting models; machine learning (ML)

1. Introduction

Building-integrated photovoltaic (BIPV) system is an advanced solar photovoltaic (PV) module designed to seamlessly integrate into building façades and roofs, enabling flexible installation across a wide range of tilt and azimuth angles to optimise solar energy capture from different orientations throughout the day [1]. In this context, the integration of BIPV systems supports urban decarbonisation strategies while simultaneously achieving environmental benefits and economic gains. The rapidly increasing investments in BIPV systems in recent years and their widespread integration into urban landscapes highlight their potential as a sustainable solution, transforming building envelopes into active renewable energy generation surfaces [2]. A key advantage of BIPV technology is its dual functionality, as it serves both as a construction material and a renewable energy generator, with a particular emphasis on harnessing the vast vertical surfaces in urban areas [2].

However, the BIPV system in particular presents a unique challenge, as power and energy generation occur across multiple orientations, making forecasting models to predict these outputs more complex [3]. For instance, PV modules facing

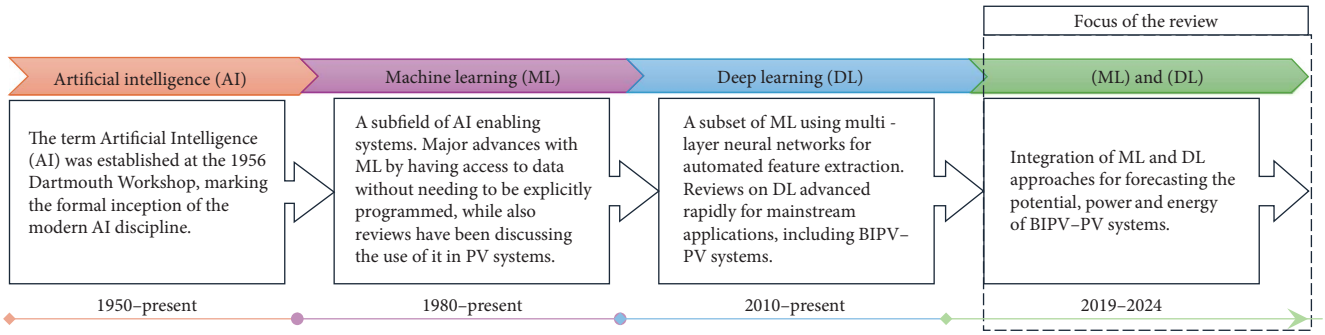


FIGURE 1: Evolution of AI towards DL applications in BIPV–PV systems.

east efficiently harness sunlight during the early morning hours, whereas west-facing modules maximise energy generation in the late afternoon [2, 3]. The integration of BIPV–PV systems into the national grid introduces specific operational characteristics that influence both power generation and demand-side dynamics [4]. Accurately predicting and analysing the output of BIPV–PV systems is therefore crucial, particularly under variable weather conditions that can significantly affect generation performance. This becomes even more critical for BIPV systems, where partial shading and intra-day fluctuations in solar energy irradiance can greatly influence both power and energy generation. Therefore, accurate forecasting under these dynamic conditions is essential for optimising energy management and supporting critical aspects, such as efficient grid integration and maintaining the reliable performance of BIPV–PV systems [5], as well as achieving numerous benefits from technical, economic and environmental conditions [6].

Consequently, the use of artificial intelligence (AI) in BIPV–PV systems is essential to provide automated, intelligent and data-driven approaches to managing complex scenarios [7, 8], as well as streamlining grid integration, supporting energy market participation and enhancing resource planning efficiency. It is important to note that these systems can supply a significant share of operational energy demand, thereby reducing both building operational costs and environmental footprint. Accordingly, to optimise and facilitate early-stage decision-making for integrating PV technology into urban planning, the majority of recent research has emphasised the need to utilise predictive tools and methodologies to evaluate the solar irradiance potential, system performance, estimated total installed capacity, output power, generated energy, environmental impacts and economic feasibility. Consequently, accurate forecasting of the BIPV–PV systems has become increasingly important for investors, developers, designers, installers, consumers and researchers [9].

As shown in Figure 1, since the early 2010s, advances in machine learning (ML) and deep learning (DL) have progressed significantly, and numerous studies have applied these techniques to forecasting methodologies. Several review papers have highlighted the crucial role of AI, particularly ML and DL, in enhancing solar energy forecasting. Here, Diagne et al. [10] presented a comparative review of forecasting methods for solar irradiance, focusing on their application in managing the integration of PV systems within small-scale insular power grids. Highlighting that forecasting global horizontal irradiance (GHI) represents the fundamental and most critical phase in the majority of PV power prediction systems. Similarly, Voyant et al. [11] reviewed ML-based methods for solar integration, while Kashyap et al. [12] reviewed and compared eight neural network architectures for solar forecasting aimed at improving prediction accuracy.

Building on this, Antonanzas et al. [13] reviewed PV power forecasting advancements, future trends and the main techniques applied in forecasting. Likewise, Barbieri et al. [14] reviewed various tools and approaches required for very short-term PV power forecasting in a large-scale grid-connected PV farm, while Das et al. [15] provided a classification-based review of the PV power forecasting models, discussing their strengths and limitations.

Additionally, it can be noted that more recent review papers have narrowed their scope to a single forecasting domain, typically by applying AI-based approaches. For instance, some reviews have focused on potential forecasting, such as predicting solar irradiance and available surface areas suitable for BIPV–PV installations [11, 16]. Similarly, others have individually reviewed the power prediction methodologies, highlighting their importance in ensuring grid stability and reliability [15, 17–21], while the remaining have concentrated on energy prediction approaches to enhance the precision of the generated energy estimates [22]. In this context, given the significant advancements in ML and DL techniques in recent years, most researchers have focused on improving the accuracy of BIPV–PV output forecasting by enhancing input data quality and employing more sophisticated, hybrid models.

Following a comprehensive systematic literature review (SLR) of research articles published over the past 5 years (2019–2024), forecasting approaches for BIPV–PV systems can be classified into three domains: potential, power and energy (PPE). The potential represents the theoretical or technical capacity of available building surfaces, for instance (façades, roofs and other building envelope elements), these surfaces are to accommodate PV modules and capture incident solar irradiance, typically determined by factors such as orientation, tilt angle, shading conditions and surface area. Power refers to the instantaneous output of the PV system at

ijer, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/er/1140262, Wiley Online Library on [20/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

a specific point in time (kW), influenced by real-time solar irradiance and system efficiency. Additionally, the energy reflects the cumulative or integrated power output over a given period (kWh) and provides a more comprehensive measure of the system's contribution to the building's energy balance or the grid over its entire lifetime. The potential prediction on its own identifies the available surface area and resource, but without integrating power and energy forecasting, it cannot capture operational behaviour or temporal variability. Likewise, power or energy prediction conducted independently of potential analysis overlooks critical spatial and architectural constraints.

Although the integration of all PPE domains into a single prediction framework is currently lacking in the literature. This review paper acknowledges the significant contributions of current individual review studies. It provides a SLR on three PPE domains for forecasting BIPV–PV systems and identifies the critical research gaps, particularly the absence of an integrated, comprehensive framework that unifies all three PPE domains. Addressing these gaps would provide a broader dimension and analytical perspective. A unified approach would enable more accurate system sizing, implementation and improved grid integration planning, ultimately enhancing the effective deployment of BIPV–PV systems and offering more robust, realistic and actionable assessments for policymakers, researchers, designers and investors.

1.1. Motivation

- Notably, there is a lack of existing review studies that simultaneously address predictions across all three domains, PPE within a single comprehensive framework.
- The aim of this review is to synthesise recent studies (2019–2024) that address all three PPE forecasting domains, rather than focusing on a single domain. This integrated approach provides deeper and more coherent insights and supports the adoption of AI-based techniques for BIPV–PV systems, while also highlighting key limitations, recommendations, current emerging trends and future directions and opportunities.

The forthcoming sections are organised in the following manner: Section 2 outlines the research methodology adopted in this review; Section 3 presents a comprehensive review of the past 5 years of research on BIPV–PV systems, including potential, power, energy and dual forecasting approaches; Section 4 examines the summary of the reviewed studies, associated limitations, emerging trends and future directions; and Section 5 provides the concluding remarks and identifies future research opportunities.

2. Materials and Methodology

2.1. Literature Search. The review method adopted in this study follows the SLR approach outlined by Xiao and Watson [23]. Conducting a systematic review enables researchers to gain a comprehensive understanding of the existing research landscape and to identify research gaps requiring further investigation. This review presents the latest forecasting and prediction methodologies applied to BIPV–PV systems over the past 5 years (2019–2024). The review adheres to the preferred reporting items for systematic reviews and meta-analyses (PRISMA) guidelines, a widely recognised and standardised framework for conducting systematic reviews. This protocol provides clear guidance for evaluating the quality and comprehensiveness of the analysis [24].

The research journal articles only presented were sourced from six well-established databases: IEEE Explore, Science Direct, Scopus, Web of Science, IEEE and the Wiley Online Library. To expand the analysis, additional manual searches were conducted through citation and reference tracking. As illustrated in Figure 2, the research process followed the methodology outlined by Moher et al. [25]. In the first stage, relevant research strings were identified, namely: (1) BIPV forecast, (2) BIPV predict, (3) building integrated photovoltaic predict* and (4) building integrated photovoltaic forecast*. The asterisk (*) was included to improve search efficiency and expand results by capturing various word forms and formats of 'predict' (e.g., prediction and predicting) and 'forecast' (e.g., forecasting and forecasted), thereby ensuring that no relevant studies are overlooked due to slight variations in wording. Once the strings were determined, they were applied systematically to each online database. The online search resulted in 282 references in the screening phase. After excluding 246 duplicated and irrelevant articles, a total of 56 articles were selected for detailed review. Of these, 36 relevant articles were identified through the database search, while 20 additional articles were obtained via reference tracking.

In order to minimise the selection bias, the following exclusion criteria were applied during the screening process: studies not directly related to BIPV–PV systems; studies lacking a forecasting or prediction component relevant to an AI-driven approach. Although the initial search returned a large number of articles, many were excluded for falling outside the scope of AI-driven forecasting. The 55 studies ultimately included in the review, therefore, reflect the specialised and emerging nature of research in this field.

3. Forecasting Methods and Performance Metrics for BIPV–PV Systems

Over the past 5 years (2019–2024), research on BIPV–PV systems forecasting has increasingly targeted specific forecasting objectives, such as potential, power or energy forecasting, to improve the accuracy of system output assessments. While several studies addressed a single prediction domain, others developed dual-domain forecasting approaches, most commonly combining potential with energy or energy with power, to enhance system forecasting robustness and practical applicability. This section provides a systematic and structured overview of each forecasting domain individually, as well as dual-domain approaches, clearly and concisely. Before

ijer, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/er/1140262, Wiley Online Library on [20/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

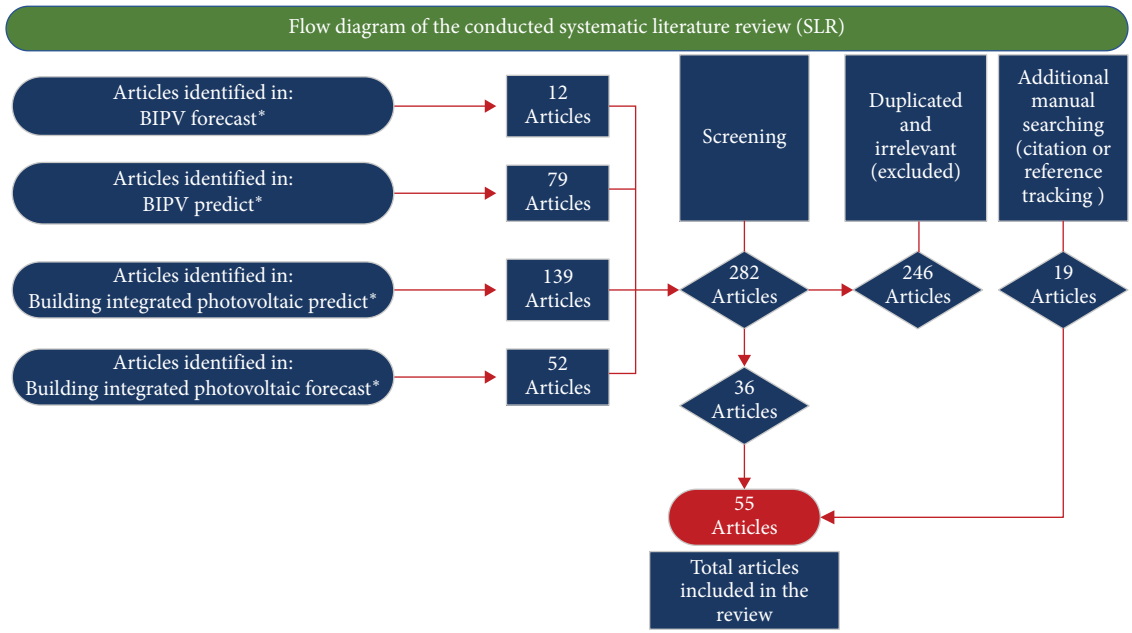


FIGURE 2: Flowchart of the review study conducted, adopted from [25].

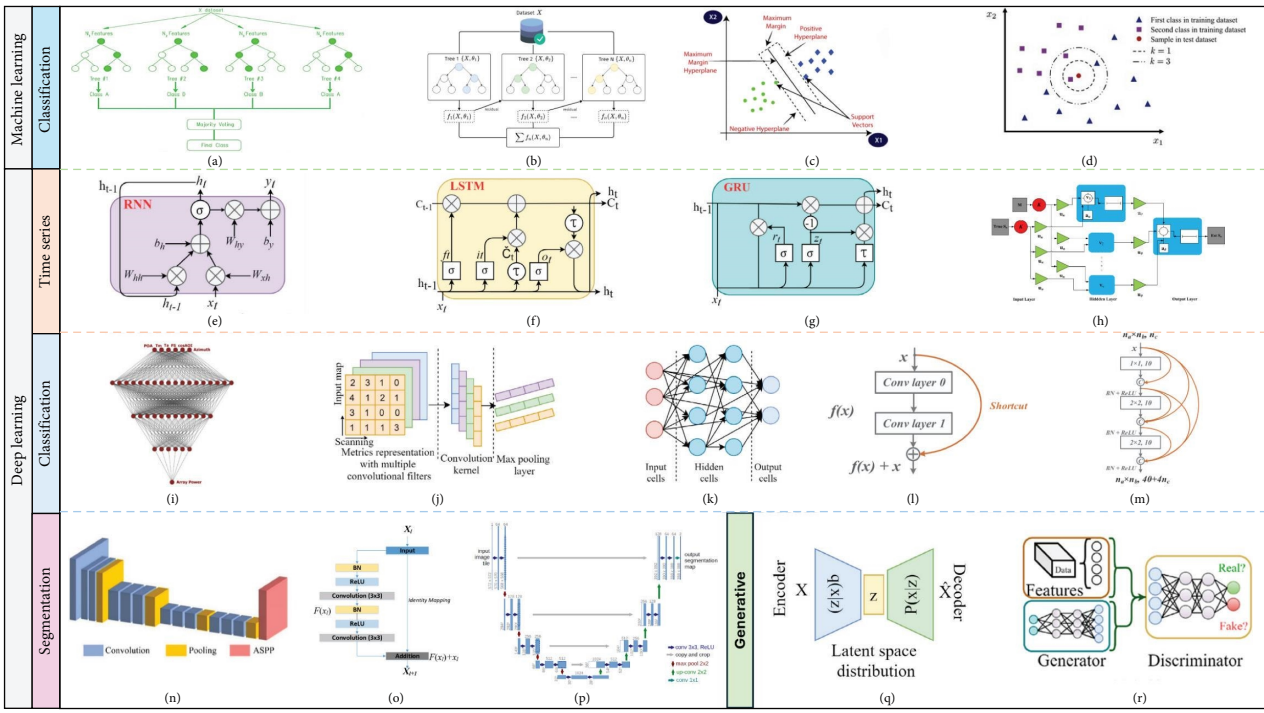


FIGURE 3: A comprehensive taxonomy of AI model architectures used in this review, diagrams were adapted from [18, 26–28]. (a) Random forest (RF). (b) XGBoost (XGB). (c) Support vector machine (SVM). (d) K-nearest neighbor (KNN). (e) Recurrent neural network (RNN). (f) Long short-term memory (LSTM). (g) Gated recurrent unit (GRU). (h) Non-linear autoregressive exogenous (NARX). (i) Artificial neural networks (ANNs). (j) Convolutional neural network (CNN). (k) Multilayer perceptron (MLP). (l) Residual network (ResNet). (m) DenseNet. (n) DeepLab-v3. (o) Residual U-Net (ResUNet). (p) U-Net. (q) Variational autoencoder (VAE). (r) Generative adversarial network (GAN).

detailing these domains, Figure 3 was developed to illustrate the key model architectures included in this review for ML and Dl models. Additionally, Tables 1 and 2 present the key definitions and performance metrics used throughout the literature to establish a common analytical basis. Table 1 summarises the main prediction indicators reported in the reviewed studies, whereas Table 2 outlines the corresponding error metrics together with their definitions.

ijer, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/er/1140262, Wiley Online Library on [20/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1: Description of prediction indicators.

Prediction indicator	Description
Number of tested methods	Refers to the overall number of various methods, techniques or algorithms that have been applied or tested in a specific study; it may include multiple approaches used to address a problem or reach the prediction's objective
Optimum algorithms/methodology used	Indicates the outperformed or most efficient algorithms or methodology chosen through testing or comparison within a study; it represents the technique that achieves the highest performance, predictive accuracy or overall effectiveness
Time horizon	Refers to the future timeframe covered by the forecasting or prediction analysis. It shows the period over which the predictions are intended to apply, such as the next minute, hour, day, month or year; forecasting weather for the next week, or planning a project over the next certain of years
Forecasting horizon	Classifies the forecasting period according to commonly used categories: very short-term (seconds to hours), short-term (hours to days), day-ahead (typically 24 h in advance) and long-term (weeks to years); this categorisation supports consistent comparison strategies across reviewed studies
Resolution	Refers to the frequency at which data points are collected, recorded or measured; common examples include minute-level, hourly, daily or other specific time intervals
Data length	Refers to the total duration or volume of data used in the study, and the period over which the data was collected; it is expressed in hours, days, weeks or months of data analysed in a time-series forecasting context.
Forecasting error metrics	Denotes the discrepancy between the actual observed values and the values predicted by a forecasting model or algorithm; it serves as a measure of the model's accuracy or effectiveness, where lower error values indicate higher precision; these errors are commonly expressed either as percentages or absolute values, such as mean absolute error (MAE) or mean squared error (MSE).

3.1. Potential Prediction. As previously defined in the introduction, the term potential refers to the theoretical or technical capacity of building surfaces or other suitable areas to harness incident solar irradiance. Accurately quantifying and forecasting this potential represents a critical first step in evaluating the feasibility and performance of BIPV–PV systems. With the continuous advancement of computational technologies, a diverse array of methodologies and analytical tools has been developed to evaluate solar irradiation across multiple spatial scales, ranging from individual buildings to broader urban contexts, thereby supporting informed decision-making in BIPV–PV systems integration. While numerous methodologies have predominantly focused on rooftop surfaces, there is a growing trend toward developing techniques that evaluate façade-based and vertical solar potential alongside rooftop installations [2]. However, the evaluation of solar irradiation on vertical surfaces within urban environments has historically faced significant challenges, primarily due to the complexity and variability of the urban landscape.

Assessing building façades requires greater analytical effort, as the intricacy of urban features, such as variations in tilt and azimuth angles, can lead to multiple irradiation peaks. These factors substantially influence the solar potential, power output and energy performance of BIPV–PV systems. Nevertheless, forecasting solar radiation on building surfaces in urban areas can be categorised into localised (micro-scale) and broad-scale (macro-scale) predictions, depending on the spatial resolution and analytical scope of the study [6]. Localised solar radiation forecasting methods primarily target the neighbourhood scale [29, 30] or the urban-block scale [31–33], and may extend to the assessment of a small number of buildings [34] or individual structures [35–37]. These approaches typically use simulated data and parametric modelling, integrating a detailed representation of city layout patterns and structural design elements. These simulations offer enhanced spatial and temporal resolution, resulting in improved predictive accuracy and more precise outcomes [38]. In contrast, broad-scale (city-wide) prediction models aim to estimate solar energy potential across entire urban areas, drawing on comprehensive building datasets [39, 40]. In this context, current methodologies for estimating rooftop solar potential at the city scale are commonly classified into four main groups: (i) three-dimensional (3D) building modelling techniques, (ii) statistical data-driven approaches, (iii) geospatial data-based methods and (iv) techniques using aerial imagery [41].

Previous studies have investigated the solar potential prediction for BIPV–PV systems across a range of application domains. AI methodologies, particularly ML and DL approaches, have demonstrated considerable potential to improve the accuracy, efficiency and scalability of solar radiation prediction models. In this context, the application of DL, particularly through artificial neural network (ANN) architectures, represents a novel and promising strategy for estimating solar radiation in complex urban environments. This approach supports the advancement of BIPV–PV systems integration within the framework of strategic urban planning and sustainable development [42]. Herein, Tehrani et al. [42] introduced a novel approach for predicting average annual solar radiation using minimal urban input data. This method integrates multiple urban parameters, including

ijer, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/er/1140262, Wiley Online Library on [20/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2: Nomenclature for forecasting error metrics.

Forecasting error metric		Brief description
CV (RMSE)	Coefficient of variation of the root mean square error	Expresses the root mean square error (RMSE) as a fraction of the mean of the observed values, enabling comparison of error magnitudes across different datasets or scales
R^2 (R-squared)	Coefficient of determination	Indicates the proportion of variance in the observed target variable that is explained by the model; values closer to 1.0 denote a better fit, meaning the model explains more variance in the data
E_{MAE}	Earth mover's mean absolute error	A variant of mean absolute error (MAE) that leverages the Earth mover's (Wasserstein) distance, quantifying the average absolute distribution shift between predicted and actual values
E_{RMSE}	Earth mover's root mean square error	Similar to EMAE, this metric is based on the root mean square error (RMSE) and uses the Earth mover's (Wasserstein) distance to evaluate the quadratic error between the predicted and actual distribution
EV score	Explained variance score	Represents the fraction of the target variance explained by the model's predictions; higher values (closer to 1) indicate that the model captures a larger proportion of the variability in the data
IoU (Jaccard index)	Intersection over union	Quantifies the overlap between predicted and true sets (e.g., areas or classes) by dividing the size of their intersection by the size of their union; it is commonly used in classification, segmentation and object detection tasks to assess the degree of spatial agreement between predication and ground truth
JSD	Jensen–Shannon divergence	A symmetric measure of difference between two probability distributions; it ranges from 0 (identical distributions) upward, with lower values indicating greater similarity between the distributions
MAD	Mean absolute deviation	Represents the mean of the absolute deviations of each prediction from the actual value; it indicates the typical magnitude of errors and is expressed in the same units as the original data
MAE	Mean absolute error	Represents the average of the absolute differences between predicted and actual values, treating all errors equally regardless of their direction; a lower MAE value indicates higher predictive accuracy
MAPE	Mean absolute percentage error	Represents the mean of the absolute percentage differences between predictions and actual values; it expresses error as a percentage of the actual values, where a lower MAPE (%) indicates higher predictive accuracy
MRE	Mean relative error	Represents the average of the errors relative to actual values, often expressed as a percentage; it indicates, on average, how large prediction errors are in relation to the true values
MSE	Mean squared error	Represents the mean of the squared differences between predicted and actual values; through squaring the errors, larger deviations are penalised more heavily, making MSE particularly sensitive to outliers
MSLE	Mean squared logarithmic error	Represents the mean of the squared differences between the logarithms of predicted and actual values; this metric mitigates the impact of large absolute differences and is useful for data exhibiting exponential growth or spanning wide value ranges
MPE	Mean percentage error	Represents the average signed percentage error of predictions; it measures the average bias in percentage terms, with positive and negative errors offsetting each other; values near zero indicate minimal systematic bias in the prediction
MPFE	Mean percent forecast error	Represents the mean of the forecast errors expressed as percentages; similar to MPE, this metric allows over-predictions and under-predictions to offset each other, thereby highlighting any overall bias in the forecasts
MedAE	Median absolute error	Represents the median of the absolute errors; this metric reflects the middle error magnitude, making it less sensitive to outliers (extremely large errors) compared to the mean absolute error
nMAE	Normalised mean absolute error	Represents the mean absolute error that has been normalised, for example, by dividing by the range or mean of the actual values; this yields a dimensionless error measure, enabling meaningful comparisons across different datasets or scales
nRMSE	Normalised root mean square error	Represents the root mean square error after normalisation, often by the mean or range of actual values, and is typically expressed as a percentage; this yields a dimensionless error, allowing model performance to be compared across datasets with different value scales
r_p (Pearson)	Pearson correlation coefficient	Represents the Pearson correlation between predicted values and actual values; it ranges from −1 to 1 and measures the strength and direction of a linear relationship, with values closer to +1 indicating a strong positive linear correlation
RMSD	Relative mean square deviation	A variant of the mean square error used as an accuracy indicator, often normalised to a reference value. In practice, it is similar to root mean square deviation (RMSE), where lower values indicate more precise model predictions
RRMSE	Relative root mean square error	Represents the RMSE expressed relative to the magnitude of the data, for example, as a percentage of the mean observed value. This normalisation allows direct comparison of prediction errors across datasets with different scales or units

ijer, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/er/1140262, Wiley Online Library on [20/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2: Continued.

Forecasting error metric		Brief description
RMSE	Root mean square error	Represents the square root of the mean of the squared differences between predicted and actual values; it provides an aggregate measure of prediction error, with larger deviations contributing more heavily due to squaring; lower RMSE values indicate higher predictive accuracy
sMAPE	Symmetric mean absolute percentage error	A modified version of MAPE that treats overestimation and underestimation equally; it calculates the average percentage error by symmetrically weighting differences, thereby reducing bias toward consistently high or low predictions
Spearman	Spearman's rank correlation coefficient	Assesses the strength and direction of a monotonic relationship between two variables (e.g., the ranks of predicted and actual values); unlike Pearson's correlation, it relies on ranked values rather than raw data, making it robust to outliers; values close to +1 indicate a strong positive monotonic relationship
WMAE	Weighted mean absolute error	A weighted extension of MAE in which each error term is multiplied by a weight reflecting the importance of corresponding data point; this approach emphasises certain errors more than others (e.g., more recent or more critical observations), providing a tailored measure of average error

Note: In general, lower values for error metrics, such as MAE, MSE and RMSE, indicate greater predictive accuracy. In contrast, higher values for variance explanation or correlation-based metrics, such as R^2, EV Score, Pearson and Spearman, reflect stronger model performance.

ijer, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/er/1140262, Wiley Online Library on [20/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

geographical coordinates (*X*, *Y* and *Z*), mean building height, occupied and unoccupied spaces and azimuth angle to improve the accuracy of solar radiation estimates across a dataset of 20 cities. Similarly, Yue et al. [6] developed an integrated method combining geographic information systems (GISs), ladybug tools (LBTs) and ML to develop a predictive model for estimating solar radiation on building surfaces at the city scale; however, their application was limited to a single urban location. In a related study, Zhang et al. [43] introduced an innovative method for generating solar irradiance time series on vertical surfaces using urban context imagery. This method emphasises computational efficiency, data precision and high spatio-temporal resolution at the urban scale. Nevertheless, a key limitation of this method lies in the long computational time required to produce fisheye images.

Li et al. [41] introduced a model, named SolarNet, designed to first extract and represent the rooftop geometry for large-scale rooftop solar potential assessment, and subsequently to perform solar potential analysis. The proposed model classifies rooftop structures into six distinct categories, including the background and multiple roof segments with their respective orientations, to support detailed and scalable solar potential estimation. Additionally, Yan et al. [44] presented a practical and efficient DL-based model for generating accurate 3D building representations to support large-scale PV potential assessments. This DL framework employs two convolutional neural networks (CNNs): the rooftop segmentation model (RSM) and the height prediction model (HPM), designed to perform rooftop segmentation and building height estimation, respectively, using high-resolution satellite imagery. However, the model exhibited reduced accuracy when applied to building façades.

A study conducted in Burgos, Spain, by Díez-Mediavilla et al. [45], evaluated the PV potential of building façades oriented toward different directions (north, south, east and west) and compared their performance to horizontal surfaces, using 45 months of daily insolation data. Measurements included GHI, diffuse horizontal insolation (DHI), beam horizontal insolation (BHI) and vertical global insolation (VGI). The results indicated that south-facing vertical façades outperformed horizontal surfaces in efficiency between October and January. Moreover, Ozoegwu [46] developed a combined approach for forecasting global solar irradiation, achieving higher accuracy and enabling long-term forecasts of upto 2 years while maintaining a typical mean percentage forecast error. Similarly, Pan and Tan [47] proposed an approach for predicting day-ahead hourly solar generation, aimed at improving forecast accuracy. As shown in Table 3, a comprehensive summary of potential forecasting studies conducted between 2019 and 2024 is presented. Evaluating potential represents a critical primary step in assessing BIPV–PV systems; however, focusing solely on potential prediction is insufficient for practical applications. To ensure effective system design and operation, it is essential to determine the power and energy outputs, as these provide a more accurate assessment of the feasibility and viability of the overall system.

3.2. Power Forecasting. In the context of BIPV–PV system performance assessment, the power output denotes the instantaneous electric power generated under specific environmental and operational conditions. This parameter serves as a fundamental indicator of system performance, and its precise estimation is essential for real-time performance monitoring and model validation. Accurate power forecasting is critical for the effective management and optimisation of solar energy systems, particularly for BIPV–PV systems, as it enables improved grid operation and enhances the efficiency of load and storage management [48]. The power output of BIPV–PV systems exhibits considerable variability, primarily due to their dependence on dynamic environmental conditions, such as solar irradiance, PV module surface temperature, and atmospheric or weather conditions. This intermittency results in fluctuating power generation. Therefore, accurate forecasting of PV power output is essential for ensuring effective real-time management of electrical power balance, particularly in grid-connected BIPV–PV systems [49]. However, the fluctuating nature of the power generated by these systems presents significant challenges for energy management systems in utilising power output predictions effectively. These predictions enhance self-consumption, reduce grid-related costs and support efficient scheduling within integrated BIPV–PV operations [50]. It is worth highlighting that previous power forecasting studies, as shown Table 4, have primarily focused on four main approaches: (1) improving prediction accuracy through hyperparameter optimisation and predictor selection; (2) evaluating the accuracy and reliability of different forecasting algorithms; (3) investigating the feasibility of hybrid models that combine multiple predictive techniques; and (4) developing prediction models that integrate real forecasted weather data [49].

In this context of forecasting models, the time horizon (or timeframe) refers to the interval over which forecasting or analytical processes are conducted. It defines how far into the future the predictions extend. The selection of an appropriate time horizon plays a critical role in forecasting, as it directly influences the accuracy, relevance and practical applicability of the results. Accordingly, the forecasting horizon may range from a few minutes, hours, a day, several months or even a year, depending on the specific application and objectives of the model. In this regard, for ultra-short-term forecasting (specifically 10 min ahead), Shin et al. [51] predicted the power output of several 3 kW coloured BIPV systems installed on walls and compared their performance using a neural-network ML-based approach, taking into account the shading losses specific to coloured BIPV systems.

Recent advancements have extended the PV power forecasting horizon to 1 h ahead, with studies introducing hybrid modelling approaches, though often without comprehensive component-level analysis. A notable study developed a dual-stream network architecture combining CNNs and gated recurrent units (GRUs), demonstrating that such hybrid architectures consistently outperform single models in forecasting accuracy [60]. Similarly, Rangelov et al. [75] evaluated three AI techniques for short-term PV power output forecasting: random forest (RF), deep neural networks (DNNs) and long short-term memory (LSTM) networks.

ijer, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/er/1140262, Wiley Online Library on [20/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 3: Potential prediction studies of the BIPV–PV system.

Model class	System type	Number of tested methods	Optimum algorithms/methodology used	Resolution	Data length	Forecasting horizon	Forecasting error	Reference
Machine learning (ML)	BIPV, PV	18	XGBoost, RR	Hourly–NA	1–14 years	Short-term to long-term	nRMSE = 8.80% nMAE = 4.06% $R^2 \leq 0.969$	[6, 47]
Deep learning (DL)	BIPV, PV	4	ANN, DNN, DGN, CNN	1 arcsecond – ~30 min	—	Short-term	MSE = 0.01 $R^2 = 0.85$ RMSE = 0.55%–1.62% JSD = 0.05–0.25 MAE = 0.05	[41–43]
Hybrid models (ML and DL)	BIPV, PV	8	RSM, HPM, NAR, ANN	10 min– monthly	3.7–14.2 years	Very short-term to long-term	MPE = 1.8%–8.1% RMSE = 15.8–25.5 MPFE = 5.67%	[44–46]

ijer, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/er/1140262, Wiley Online Library on [20/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 4: Power forecasting studies for the BIPV–PV system.

Model class	System type	Number of tested algorithms	Optimum algorithms	Time horizon	Resolution	Data length	Forecasting horizon	Forecasting error	References
Machine learning (ML)	BIPV	2	XGBoost, RF	1 h	1 h	2 years	Very short-term	MAE = 30% (E) 40% (S); R^2 = 0.60–0.69	[48]
	BIPV	1	NN	10 min	0.5 month	6.5 months	Very short-term	R^2 = 0.41 (train), = 0.38 (test)	[51]
	PV	24	KR, MLP	Day-ahead	15 min	2 years	Short-term	RMSE = 13.9%	[52]
	PV	12	RF	Day-ahead	1 h	3 years	Short-term	MAE = 6.13%	[53]
	PV	1	KNN	1 h-ahead	1 h	1.35 years	Very short-term	MAPE = 13%	[54]
	PV	3	bSBFM	Day-ahead	5 min	2 years	Short-term	nRMSE = 15.3% MRE = 10.8%	[55]
	PV	10	AutoML (indicator-AML)	Day-ahead	30 min	2 years	Short-term	MAE = 4.27 RMSE = 7.77% R^2 = 94.8%	[56]
Deep learning (DL)	BIPV	3	Hybrid DL (LSTM–DF–FF)	Day-to-day	60 min	2022–2023	Short-term	MSE = 19.54 RMSE = 2.42 R^2 = 0.88 MAE = 3.5 MAPE = 0.875	[57]
	BIPV	1	ANN	3 years	5 min	2 years	Long-term	MRE = 6%–15.9%	[5]
	BIPV	3	ANN	Day-ahead	5 min	6 months	Short-term	RMSE = 4.42 R^2 = 0.88 MAE = 3.0 MAPE = 0.55	[58]
	BIPV	3	EN, FFN, GRN	Day-ahead	—	3 years	Short-term	RMSE = 0.285–0.426 MAPE = 0.189–0.241	[59]
	PV	2	Dual-stream (CNN–GRU)	1 h-ahead	5 min	—	Very short-term	MAE = 0.0106 RMSE = 0.0896 MSE = 0.0080	[60]
	PV	2	1D-CNN, LSTM	15–120 min	15 min	2 years	Very short-term	–CNN:MAPE = 49%–70% –LSTM:MAPE = 58%–5%	[61]
	PV	1	NARX	Day-ahead	1 h	180 days	Short-term	R^2 = 0.88 MAPE = 24.21%	[49]
	PV	1	Rec-LSTM	24 h	15 min	1 year	Short-term	nRMSE = 14.47%	[62]
	PV	1	Residual UNet	—	5 min	—	Very short-term	RMSE = 4.50% (clear) and 11.29% (overcast)	[63]
	PV	5	GCNs	—	5 min	—	Very short-term	MAE = 0.177 RMSE = 0.336	[64]
	PV	4	GRUPs	Day-ahead	5 min	3 years	Short-term	nMAE = 4.12% nRMSE = 6.83%	[65]
	PV	6	ANN	Day-ahead	15 min	3 years	Short-term	MAE = 24.18 RMSE = 46.86	[66]
	PV	1	LSTM	24 h-ahead	5 min	3.8–11 months	Short-term	MASE > 1.0 (cold), ≈0.9 (warm)	[50]
	PV	1	Modified LSTM–RNN	Day-ahead	15 min	5 years	Short-term	RMSE = 5.68% MAE = 2.35%	[4]

ijer, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/er/1140262, Wiley Online Library on [20/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 4: Continued.

Model class	System type	Number of tested algorithms	Optimum algorithms	Time horizon	Resolution	Data length	Forecasting horizon	Forecasting error	References
	PV	3	DenseNet	—	1 h	—	Short-term	MAE = 0.152 MSE = 0.081	[67]
	PV	6	RNN	64 days	3 h	1.2 years	Long-term	CV (RMSE) = 0.51 MAD = 1.39 MAPE = 23.79%	[68]
	PV	1	DCNN	Day-ahead	1 min	5 days	Short-term	MAE = 109.48 RMSE = 163.15	[69]
	PV	3	DRNN, LSTM	Day-ahead	1 h	1 month	Short-term	RMSE = 7.54 MAE = 4.37 MAPE = 15.87%	[70]
	PV	4	ANN	Hourly	—	2 years	Short-term	RMSE = 2.14 MSE = 4.60 MAE = 2.13	[71]
Hybrid models (ML and DL)	BIPV	3	OLM	4 days	15 min	121 days	Short-term	RMSE = 2.24 kW MAE = 1.12 kW WAPE = 4.66%	[3]
	BIPV	3	CEEMDAN, RF, LSTM	—	—	—	—	RMSE = 2.97	[72]
	BIPV	1	CSO–GWO–LSTM	Hourly to monthly	—	1 year	Very short-term to short-term	MSE = 0.0036–0.151 (daily–monthly)	[73]
	PV	5	SVMD–SE– SSA IMSOA	—	—	—	—	sMAPE = 4.78%, 4.76% and 6.03%	[74]
	PV	3	RF, DNN, LSTM	1 h-ahead	1 h	—	Very short-term	MAE = 309.46 W; RMSE = 443.42 W	[75]
	PV	3	COA, CNN, LSTM	—	15 min	—	Very short-term	RMSE = 0.039 R^2 = 0.983 MAE = 0.021	[76]
	PV	2	RCC, RCC–LSTM	—	5 min	—	Very short-term	MAE = 0.587 RMSE = 0.94	[77]
	PV	3	SDA–GA–ELM	—	5 min	—	Very short-term	RMSE = 0.2107 MAE = 0.2367	[78]
	PV	2	Modified vs. traditional model	1 day	15 min	4.5 years	Short-term	E_{RMSE} = 8.41 E_{MAE} = 7.67	[79]
	PV	3	HIMVO–SVM	—	30 min	—	—	MAE = 0.2805	[80]
	PV	2	Weather clustering, ANN	—	1 h	—	Short-term	WMAE: 1%–30.39% (sunny) 9%–750% (cloudy)	[81]
	PV	3	Ensemble model	1 month	5 min	~5 years	Long-term	RMSE = 0.147 MAE = 0.0985 MAPE = 0.912	[82]

ijer, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/er/1140262, Wiley Online Library on [20/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

The study concluded that both the RF and DNN models achieved accurate solar power forecasts and effectively captured sudden fluctuations, while the LSTM model exhibited comparatively weaker performance in predicting PV power generation.

Furthermore, Polo et al. [48] examined hourly power forecasting for BIPV systems deployed on vertical façades, analysing two strings oriented towards the south and east. Their approach utilised decision-tree ML-based models. The results confirmed the superior performance of algorithms, such as RF and gradient boosting methods, including XGBoost, for time-series forecasting of BIPV output. Likewise, Eroshenko et al. [54] proposed a forecasting model designed for a 15 MWp installed capacity, addressing the need for accurate and reliable power generation forecasts under challenging weather conditions. The k-nearest neighbours (KNNs) algorithm was identified as the most suitable technique for preliminary data filtering, employing clustering methods to detect results consistent with recurring weather and seasonal patterns.

Additionally, Khandakar et al. [71] estimated PV output power through ANNs and compared the results with several regression-based approaches. The ANN model demonstrated superior performance relative to conventional regression techniques, such as linear regression, decision trees, and gaussian process regression (GPR). Whereas, Bellagarda et al. [61] developed a short-term forecasting model with a 2-h prediction horizon, employing ANNs in data-limited environments. The study integrated two ANN variants (GRU and one-dimensional CNN) and applied diverse transfer learning (TL) strategies to enhance ANN performance using the remaining subset of the restricted PV dataset.

It is worth noting that the majority of power forecasting studies over the past 5 years (2019–2024) have predominantly focused on the day-ahead prediction horizon. In this context, kumar et al. [57] proposed an innovative neural-network-based approach to estimate the power output performance of BIPV systems operating under warm and moisture-rich conditions. The method relies on meteorological inputs such as solar irradiance, ambient temperature, and wind speed. It integrates a novel DL architecture employing LSTM networks, with optimised feature selection achieved through dragonfly (DF) and firefly (FF) algorithms, resulting in a hybrid learning framework.

In addition, Jeong et al. [49] proposed a feasible method for hourly day-ahead BIPV power forecasting using a simple meteorological prediction system that does not include solar irradiance forecasts. The study introduced a novel feature extraction process to identify on-site hourly sky conditions from weather data, thereby enhancing the estimation of solar irradiance on PV panels. This approach significantly improved short-term BIPV power output forecasts. However, the non-linear autoregressive exogenous (NARX) model architecture, belonging to the recurrent neural network (RNN) family, performed well under clear and partially cloudy conditions but exhibited reduced accuracy during overcast and cloudy periods.

Whereas Markovics and Mayer [52] conducted a comparative assessment of twenty-four ML algorithms applied to deterministic next-day power prediction based on numerical weather prediction (NWP) data, the study evaluated these models using 2 years of measurements recorded at 15-min intervals from 16 PV facilities located in Hungary. The findings revealed that the kernel ridge regression (KRR) and multilayer perceptron (MLP) models demonstrated the highest predictive accuracy, achieving a forecast skill score upto 44.6% higher than persistence. Although the KRR model was the most accurate overall, it exhibited considerable computational time demands and substantial memory consumption. In contrast, the MLP, a fundamental architecture within the feedforward neural network family, delivered comparable accuracy with significantly shorter training times, making it a more practical option for real-world deployment. Additionally, Li et al. [62] introduced a comprehensive probabilistic forecasting framework for PV power output that was robust to missing data. Using historical PV generation records as input variables, the model was based on a recursive LSTM (Rec-LSTM) network capable of generating probabilistic forecasts across multiple future time steps. This method effectively handled missing data scenarios during both offline and online operation stages. Furthermore, Kabilan et al. [58] proposed an ML-based prediction model for BIPV systems, accounting for the different PV orientations. This model utilised data science tools, including a preprocessing pipeline involving data validation, algorithm selection, meteorological data grouping and performance evaluation. In a subsequent study, Kaliappan et al. [59] optimised and forecasted the performance of building-integrated semi-transparent photovoltaic (BISTPV) systems, comparing three ANN models for performance prediction. Similarly, Maitanova et al. [50] developed a fully automated forecasting method for estimating PV power generation over a 24-h horizon, applied to two installed systems differing in capacity and location (1.14 kWp in Oldenburg and 99.9 kWp in Munich). The prediction employed an LSTM architecture. In the same context, Wang et al. [4] introduced a day-ahead DL forecasting approach employing LSTM-based RNNs with temporal correlation mechanisms, integrating time-dependent features such as annual recurrence and neighbouring-time similarity. A complementary partial daily pattern prediction (PDPP) framework was also proposed to enhance the forecasting accuracy.

Moreover, Sangrody et al. [55] predicted the next day's PV power output using similarity-based forecasting models (SBFMs) across three configurations: basic, categorical and hierarchical. These models were tested under three different metrological data availability scenarios, using inputs sampled at 5-min intervals with either one or a limited number of weather variables. The results, obtained from a decentralised PV system, indicated that the Basic SBFM (bSBFM) offered a simpler, more effective, and higher-accuracy alternative to conventional baseline models. Likewise, Huang and Kuo [69] introduced an advanced DNN architecture, referred to as PVPNet, for day-ahead PV power forecasting. The model was validated using empirical datasets, demonstrating that the predicted outputs closely matched the observed measurements. Furthermore, Wen et al. [70] developed a deep RNN (DRNN) including LSTM units to forecast short-term PV power. The proposed model outperformed both the MLP and support vector machine (SVM) models.

ijer, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/er/1140262, Wiley Online Library on [20/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Considering longer-term forecasting horizons beyond day-ahead, Dimd et al. [3] predicted the BIPV system output power of 181.15 kWp for the subsequent 4 days using three models and examined the effect of mixed orientations (tilt and azimuth angles) across six configurations, using an LSTM model. Likewise, Ragupathi and Ramasubbu [73] achieved weekly forecasting by developing and analysing BIPV prediction algorithms across multiple forecasting horizons, extending to week-ahead intervals to support grid management applications. The proposed multitask learning framework based on LSTM neural networks was implemented and evaluated across these forecasting horizons.

Furthermore, when extending the forecasting horizon to approximately 2 months, Lee et al. [68] introduced a novel approach to enhance forecasting of power output by combining input variable transformation with ML techniques. In this study, six ML algorithms were evaluated to identify an optimal framework capable of hourly power forecasting using localised meteorological forecast data. The results showed that the RNN achieved superior performance compared with the five alternative models. Additionally, feature engineering was implemented by integrating dropout-based observational data into the standardised sky index during both training and forecasting stages, resulting in a 20% improvement in hourly forecast accuracy. The forecasting horizon was further extended to 2 years ahead by Polo et al. [5], who developed ANN to simulate BIPV power generation for five small-scale systems at 5-min intervals, using on-site meteorological and solar irradiance data, along with shadow distributions estimated from digital surface model (DSM) as input variables.

It is important to highlight that the advancement of AI in power forecasting has substantially improved forecasting accuracy on multiple timescales. However, persistent challenges remain in addressing uncertainties arising from weather variability, shading effects and system degradation, all of which complicate the refinement of predictive models for BIPV–PV systems. Meanwhile, a comprehensive understanding of energy forecasting remains crucial to unlocking the applicability of BIPV–PV forecasting, maintaining grid stability and ensuring sustainable energy resource management.

3.3. Energy Prediction. Within the framework of long-term BIPV–PV performance evaluation, energy output refers to the cumulative electrical energy generated over a defined time horizon or throughout the system's operational lifetime. Effective building energy management systems (BEMSs) require a dependable approach for predicting future energy demands in conjunction with the deployment of renewable energy technologies. In this context, forecasting BIPV–PV energy output is essential for the effective management of electric energy generation and storage processes within the electric grid on a daily or hourly basis. Accurate forecasts enable optimised grid operations, ensuring a reliable supply of energy and better seamless integration of renewable energy systems into grid infrastructure [83]. Herein, predictive modelling is therefore instrumental in facilitating efficient and reliable operation across the energy sector. Energy forecasting, in this regard, includes various aspects of the energy landscape, with the primary focus on load forecasting at both grid and individual building scales [84]. It is crucial to recognise that precise energy forecasting is crucial for BIPV–PV systems, as it enables their effective involvement in energy markets while enhancing strategic resource allocation [83].

It can be noted that there was a lake of energy forecasting studies, although Abouelaziz and Jouane [85] introduced an innovative methodology that combines photogrammetric analysis with advanced DL models was introduced to address the challenge of BIPV decarbonisation and to enhance the precision and robustness of PV output forecasting using AI approaches. The findings validate the CNN model's capability in accurately forecasting output energy, as evidenced by its strong agreement between predicted and actual outputs. Additionally, Jasinski et al. [86] conducted a day-ahead energy forecasting for a 15 kWp system, developing a feedforward ANN model that employed historical operational data and meteorological variables from comparable previous days, supplemented with forecast data for the target prediction day. Two forecasting strategies were compared (panel-level and system-level) for output prediction. Furthermore, an LSTM model was employed to validate the outcomes of the feedforward ANN, confirming its reliability. It has been observed that most existing studies in energy forecasting, as shown in Table 5, tend to focus on DL-based approaches, due to their ability to model non-linear interactions, capture temporal dependencies and long-range correlations that characterise BIPV–PV energy generation behaviour. However, integrated methods that combine multiple forecasting domains play a critical role in developing a comprehensive understanding of the operational characteristics of BIPV–PV systems. Recent dual forecasting models have successfully integrated two of the three PPE domains, demonstrating the advantages of multi-domain forecasting for enhanced predictive performance.

3.4. Dual Forecasting. Dual prediction refers to integrated approaches that combine multiple forecasting domains, these domains aim to develop a comprehensive understanding of the operational characteristics of BIPV–PV systems. Recent hybrid models have successfully merged two of the three key PPE forecasting domains, dominating the advantages of multidimensional modelling for improved predictive accuracy and system performance. Current studies on dual forecasting, as shown in Table 6, have primarily concentrated on two combined domains of system design prediction, namely potential and energy, power and energy, as well as potential and power. These studies aim to achieve a broader analytical scope and optimise the precision and performance through integrating advanced methodologies, such as ML and DL algorithms.

Zhao et al. [2] developed two predictive models addressing both potential and energy, introducing a method to forecast wide-area solar exposure and energy production on architectural surfaces within ML frameworks. Similarly, Ni et al. [87] introduced an approach to forecast PV potential

ijer, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/er/1140262, Wiley Online Library on [20/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 5: Energy forecasting for the BIPV–PV systems.

Model class	System type	Methods tested	Algorithms/ methodology	Time horizon	Resolution	Data length	Forecasting horizon	Forecasting error	References
	BIPV	7	CNN	—	—	—	Very short-term	Spearman = 97% $r_p = 98\%$ $R^2 = 96\%$	[85]
Deep learning (DL)	PV	1	ANN	Day-ahead	10 min	9 months	Short-term	MAPE:16.93% (3-phase); 9.37% (single-phase)	[86]
	PV	1	MLP	Day-ahead	Hourly	2.9 years	Short-term	MAE = 6.697 RMSE = 13.260	[84]

and energy generation using DL and GIS techniques, while accounting for obstacles and shading effects. The DL approach, specifically employing DeeLab-v3, was used to identify rooftop PV systems installed across Shanghai. Additionally, Fara et al. [83] carried out power and energy forecasting using two models: the autoregressive integrated moving average (ARIMA) and ANN. These models were applied to forecast solar radiation potential and predict energy production for two case studies, namely a laboratory-scale BIPV system and a utility-scale PV plant. The study concluded that the ARIMA was more efficient and outperformed the ANN model.

Furthermore, Zhen et al. [88] proposed a hybrid mapping model which combines multiple DL techniques, including CNN, LSTM and ANN, to predict the prospective levels of solar radiation and electrical PV power output. The model used sky image data, which are first clustered using feature extraction through a convolutional autoencoder, followed by the application of the K-means clustering algorithm. Similarly, Cui et al. [79] investigated the forecasting of solar irradiance and power using two models (modified and traditional). The results indicated that the enhanced physical modelling approach provided superior predictive accuracy compared with conventional techniques and demonstrated greater robustness under varying meteorological conditions.

It is noteworthy that the hybrid prediction methods represent a significant advancement towards the next generation of BIPV–PV forecasting frameworks. The integration of multiple forecasting domains and learning algorithms enables the development of adaptive, data-driven models capable of supporting real-time optimisation of PPE forecasting. Nevertheless, continued research in this field remains crucial to further enhance model generalisation and robustness under diverse environmental and operational conditions.

4. Discussion

The systematic review covering 2019–2024 highlights a clear evolution in BIPV–PV forecasting methodologies, progressing from task-specific ML applications towards hybrid modelling approaches capable of simultaneously addressing spatial segmentation, time-series forecasting. These approaches now extend from solar irradiation and available area estimation for potential prediction to power output and energy yield forecasting. This transition reflects a continuing shift from single-domain to dual-domain forecasting, with a notable trajectory observed towards the unified hybrid PPE domains identified in this review, where roof/façade segmentation is coupled with time-series learning and long-term power energy yield forecasting.

The review also highlights the forecasting error metrics most commonly used in the literature. Although recent research progress has improved predictive accuracy, the inconsistent use of diverse error metrics limits unified comparative assessment across methodologies. Table 7 summarises the current PPE forecasting domains, the commonly used techniques, key performance outcomes, metric purpose, forecast range and associated strengths, limitations and applicability.

4.1. Limitations. Various studies demonstrate practical limitations that constrain the deployment of current AI-based approaches. For instance, Zhang et al. [43] proposed a method that, although conceptually robust, requires a substantial computational time due to its high model complexity. Similarly, the model developed by Yan et al. [44] showed a reduced predictive accuracy when applied to building façades, reflecting the increased geometric variability and annotation difficulty associated with façade datasets. These findings indicate the need for lightweight model architectures, alongside façade-specific data augmentation and calibration strategies. A major limitation is the limited ability of current models to generalise across facades that vary in geometry, material properties, shading characteristics and urban context.

Additionally, while several models performed strongly under stable conditions, they showed limitations when evaluated under real-world constraints. The NARX model, for instance, experienced reduced accuracy during overcast or cloudy periods, whereas KR regression, despite delivering high predictive accuracy, required significant computational time and memory resources, limiting its feasibility for real-time or embedded deployment.

An evident limitation across the reviewed studies is the lack of long-term power and energy forecasting. The longest prediction model for power forecasting extended to approximately 3 years, which contrasts with the typical 25–30-year operational life span of BIPV–PV systems. The reliance on historical, data-driven patterns constrains the ability of

ijer, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/er/1140262, Wiley Online Library on [20/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 6: Dual forecasting of the BIPV–PV systems.

Model class	System type	Forecasting domain	Number of tested algorithms	Optimum algorithms	Time horizon	Resolution	Data length	Forecasting horizon	Forecasting error	References
Machine learning (ML)	BIPV	Potential and energy	24	RF	—	—	—	Short-term	RMSE = 2.555 MSE = 6.530 $R^2 = 0.972$ MAE = 1.175	[2]
	BIPV	Power and energy	2	ARIMA, ANN	15 min–10 days	—	2–10 days	Short-term	$R^2 = 0.811$	[83]
Deep learning (DL)	PV	Potential and energy	2	DNN + GIS tools	—	—	—	Short-term	IoU = 0.84 and 0.09	[87]
Hybrid models (ML and DL)	PV	Potential and power	1	ANN, CNN, LSTM	1 h	15 min	1 year	Very short-term	RMS = 107.7 MAE = 65.7	[88]
	PV	Potential and power	2	Modified ANN	1 day	15 min	4.5 years	Short-term	ERMSE = 8.41 EMAE = 7.67	[79]

ijer, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/er/1140262, Wiley Online Library on [20/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 7: Comparative assessment demonstrating the evolution in BIPV–PV systems forecasting approaches.

	Forecasting domain	Model class	Best performing techniques (from Tables 3–6)	Performance outcome	Common error metrics	Metric purpose	Forecast range	Strengths	Limitation	Applicability
Focus of the review (2019–2024)	Potential (façade/ roof solar potential)	ML/DL/ hybrid	– XGBoost – RR – ANN – CNN – DGN – DNN	Stable performance for irradiance regression and spatial mapping	– nRMSE – nMAE – MSE – RMSE – R^2	Evaluating regression accuracy and segmentation performance.	Short to long-term	Predict façade/ rooftop PV potential with limited data	Segmentation performance decreases in shaded and complex façades	Urban mapping, city-scale PV planning
	Power (instantaneous BIPV–PV output)	ML/DL/ hybrid	– XGBoost – RF – KNN – CNN–GRU – LSTM–RNN – ANN–CNN–LSTM	Highest accuracy when recurrent DL and combined models are applied	– MAE – RMSE – R^2 – MAPE – rRMSE – RMS – MSE	Measures error magnitude, relative error and temporal prediction precision	Very short to short-term	Captures intermittent power fluctuations, shading effects, and rapid PV output dynamics	Requires large, labelled datasets, performance degrades under rapid cloud transients	Smart inverter control, grid scheduling, BIPV system operation
	Energy (daily/ short-term energy prediction)	DL (primary)	– CNN – ANN – MLP	High correlation and precision in cumulative energy yield	– MAE – MAPE – RMSE – R^2	Evaluates relative yield accuracy and energy colouration performance	Very short to short-term	High precision for non-linear relationships and cumulative energy patterns	Lacks in lifetime yield prediction, accuracy declines under variable operating conditions	Day-ahead consumption matching, energy output forecasting
	Dual forecasting (potential-energy, or power–energy)	ML/DL/ hybrid	– RF – DNN–GIS – ARIMA–ANN – DNN–GIS tools – ANN–CNN–LSTM	Balanced regression performance, moderately stable in coupled forecasting domains	– RMSE – MAE – R^2 – rRMSE – IoU	Capture coupled regression behaviour, intermittency patterns and spatial assessment	Short-term	Useful in preliminary forecasting phases where special and temporal coupling is required	ARIMA-ANN unstable under dynamic clouds, limited system integration insight	Initial system evaluation, foundation for advanced future PPE forecasting studies
Proposed future work	Hybrid PPE forecasting (unified potential, power and energy)	Hybrid (ML and DL)	– Segmentation models (GANs/SAM) – RF–LSTM – ANN–CNN–LSTM – Modified ANN – TL – GNNs – RL – GNNs	Most robust overall, strongest cross-domain performance when segmentation and time-series learning are integrated	– IoU – R^2 – MAE – RMSE	Measures PPE consistency, multi-class façade, power and energy predictive performance	Short-term to long-term	Multi-stage learning reduces error by combining spatial segmentation with temporal forecasting	Requires large datasets	Building Façade/ roof segmentation, day-ahead power forecasting with consumption alignment and long-term/lifetime PV energy yield forecasting

ijer, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/er/1140262, Wiley Online Library on [20/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

current AI models to produce reliable long-term forecasts. As a result, existing approaches remain insufficient for life-cycle planning, economic evaluation and policy compliance. A notable gap identified across the reviewed studies is the limited integration of economic metrics into forecasting frameworks. Although technical prediction remains the dominant focus, current studies lack the inclusion of financial indicators such as levelised cost of electricity (LCOE), capital and operational expenditure or payback period. This limitation restricts the applicability of current forecasting approaches for investment planning and policy evaluation.

4.2. Recommendations. The reviewed studies of AI-based BIPV–PV systems models remain limited by their dependence on biased training data, limited generalisation across diverse urban environments, high computational cost and sensitivity to lighting and weather conditions. These limitations highlight the need for further systematic validation using real-world façade datasets to ensure reliable deployment. Consequently, there is an expanding requirement for forecasting approaches that balance accuracy with practicality, including lightweight CNN architectures that have demonstrated improved performance over which outperforms the current simulation tools [85].

Building on these considerations, advanced AI methods, including supervised ML and ANNs, combined with heuristic optimisation algorithms [89], should be employed to develop robust data-driven models for predicting on-site renewable power generation and to support structured uncertainty and sensitivity analyses [90], which are essential for reliable BIPV–PV performance forecasting. Furthermore, hybrid approaches that integrate LSTM architectures with RNN can enhance temporal forecasting performance, as they process sequential measurements, update the RNN state, and iterate across temporal sequences [91].

A promising direction for improving building façade segmentation involves adapting multi-stage computer vision (CV) methods that integrate large-scale vision models. Approaches that combine DINO-based object detection, SAM-guided segmentation and stable diffusion-based inpainting have the potential to automatically remove occlusions, align facades and generate a building façade mask. This approach would provide a scalable and more reliable foundation for subsequent BIPV–PV performance estimation models [92].

Furthermore, to maximise the performance of BIPV–PV systems, future work integrating AI approaches into techno-economic PPE forecasting should prioritise the PV module degradation factor, accurate system sizing [93], consider the effect of orientation, tilt and azimuth angles [94], and optimise the power sizing ratio (PSR) [95], while thoroughly characterising system losses, inverter efficiency and real-world operating behaviour based on using high-quality measurement data. Given the specific thermal and associated shading conditions with the BIPV integration, detailed consideration of façade-specific mismatch losses, elevated operating temperatures and urban-environment shading effects is crucial to ensure realistic performance optimisation. Moreover, an integrated assessment of the solar radiation potential across all relevant building surfaces is necessary to determine the most effective BIPV–PV system layouts and to identify opportunities for power output optimisation [96]. The interaction between BIPV systems, heating, ventilation and air conditioning (HVAC) systems and the thermal properties of building envelopes must be systematically evaluated to balance indoor thermal comfort requirements while minimising overall energy consumption [97].

4.3. Emerging Trends and Future Directions. TL has gained increased recognition and emerged as a promising approach for adapting pre-trained models developed from data-rich BIPV–PV systems to new environments with limited data availability, thereby improving forecasting accuracy and reducing both data requirements and computational time and costs [98, 99]. Given the dynamic operational nature of BIPV systems, reinforcement learning (RL), which is an ML paradigm for sequential decision-making, offers a promising solution due to its ability to adapt to continuous operational changes. RL learns optimal scheduling strategies through direct interaction with the operational environment, rather than relying on labelled data [100]. Herein, RL represents a demanding direction for future intelligent BIPV systems, supporting real-time adaptability for power and energy demand forecasting, as well as system design and configuration [100, 101].

Another emerging development in PV power forecasting is the use of graph neural networks (GNNs), a type of DL model capable of capturing the irregular relationships between different power plants and accounting for the impact of cloud movements and weather variability on power generation [102]. GNNs have also been applied to time-series forecasting tasks through the construction of time series graphs, provided that the data exhibited an underlying graph structure [103]. Furthermore, physics-informed ML (PIML) is another promising trend that combines data-driven ML techniques with fundamental physical laws. This synergy enables forecasting models to benefit simultaneously from data-based learning and physics-based knowledge, thereby supporting more rational model behaviour and reducing forecasting error [104, 105].

Future BIPV–PV forecasting models should reduce dependence on high-quality training data, improve generalisation across different environments [106], lower computational cost and mitigate sensitivity to lighting and weather variability [107]. The observed challenges, including long processing times for image-based models and reduced façade prediction accuracy, highlight the need to improve model efficiency specifically for BIPV applications. Consequently, future research should prioritise the development of facade-oriented training datasets and generalised frameworks capable of handling diverse BIPV integration surfaces. The identified operational weaknesses in existing models, such as NARX and KR regression, further emphasise the importance of computationally efficient architectures and hybrid frameworks capable of maintaining accuracy under variable weather conditions without necessitating excessive processing requirements.

ijer, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/er/1140262, Wiley Online Library on [20/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

The review identifies the limited prediction horizon of current models as a critical weakness, highlighting the need for forecasting approaches capable of generating reliable multi-decade energy yield forecasts, considering the annual degradation and maintenance records consistent with typical PV lifetimes. Therefore, future research should focus on hybrid frameworks combining AI-based forecasting with long-term degradation modelling and climate projection inputs, enabling robust estimates of total energy yield over the BIPV–PV system's full operational lifetime.

Future research should also support detailed techno-economic assessments, enabling the formulation of practical policy pathways to accelerate the integration of BIPV–PV systems within urban landscapes. Although the maximum forecasting horizon has been extended from minutes to 3 years, this remains insufficient given the typical 25-year operational lifetime of BIPV–PV systems. Additionally, the economic parameters are largely absent from existing forecasting models, despite their essential role in assessing overall project viability. Employing the potential of advanced DL and hybrid methodologies, cities can strategically optimise their solar potential, where models that consider building height, shading constraints, façade availability and urban geometry offer faster and accurate solar predictions.

5. Conclusion

This systematic review investigates the recent advances of forecasting methodologies for BIPV–PV systems from 2019–2024 and categorises them into three domains: PPE. It identifies a marked progression from domain-specific models to integrated hybrid forecasting domains. A promising direction in BIPV–PV forecasting is the use of multiple algorithms to improve forecasting precision and achieve accurate results, reflecting a growing shift towards multi-stage model predictive frameworks.

Hybrid models, such as the combinations of ANN and CNN, have demonstrated notable performance in forecasting power and energy outputs by exploiting the corresponding strengths of both modelling approaches. DGNs are often built on either CNNs or RNNs architectures, which have shown significant effectiveness in handling intricate, high-dimensional datasets, including aerial visual inputs and temporal sequences, thus making them highly effective for solar radiation potential prediction. Similarly, ANNs and SVMs are commonly employed data-driven techniques for sequential temporal datasets, solar irradiance modelling, due to their ability to capture non-linear structures and complex behaviour trends within the data. Despite notable advancements, existing approaches exhibit key limitations, including inconsistent use of evaluation metrics, high computational demand, reduced façade segmentation accuracy due to geometric variability, weather-sensitive predictive performance, and the absence of long-term degradation energy forecasting. These constraints limit the suitability of current models for real-world deployment.

The review identified several promising approaches for improving forecasting reliability, including lightweight model architectures, façade-specific data augmentation and systematic validation using real-world building datasets. Additionally, several emerging trends (TL, RL, GNNs and PIML) offer an opportunity to advance the BIPV–PV forecasting and provide scalable pathways to reduce data dependences, improve generalisation, enhance computational efficiency and lower computation cost.

Future studies should prioritise the development of façade datasets, robust hybrid frameworks capable of operating reliably under variable weather, shading conditions and multi-decade forecasting models. The review highlights the limited integration of economic parameters in most forecasting studies, emphasising the need for integrated techno-economic models that can inform both performance and financial decision-making.

Nomenclature

AI:	Artificial intelligence
AML:	Applied machine learning
AutoML:	Automated machine learning
ANN:	Artificial neural network
ARIMA:	Autoregressive integrated moving average
BEMS:	Building energy management systems
BHI:	Beam horizontal insolation
BIPV:	Building-integrated photovoltaic
BISTPV:	Building-integrated semi-transparent photovoltaic
bSBFM:	Basic similarity-based forecasting model
CEEMDAN:	Complete ensemble empirical mode decomposition with adaptive noise
CNN:	Convolutional neural network
COA:	Cuckoo optimisation algorithm
CSO:	Cat swarm optimisation
CV:	Not specified
DGN:	Deep generative network
DHI:	Diffuse horizontal insolation
DL:	Deep learning
DNN:	Deep neural network
DRNN:	Deep recurrent neural network
DSM:	Digital surface model
E_{MAE}:	Earth mover's mean absolute error
E_{RMSE}:	Earth mover's root mean square error
EV:	Score explained variance score
FF:	Firefly algorithm
FFN:	Feed forward neural network
GA–ELM:	Genetic algorithm–optimised extreme learning machine
GHI:	Global horizontal irradiance
GIS:	Geographic information systems
GNN/GNNs:	Graph neural networks
GPR:	Gaussian process regression
GRN:	Generalised regression neural network

ijer, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/er/1140262, Wiley Online Library on [20/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

GRU:	Gated recurrent unit
GRUPs:	Gated recurrent unit pools
HIMVO:	Hybrid improved multi-verse optimiser
HPM:	Height prediction model
IoU:	Intersection over union
JSD:	Jensen–Shannon divergence
KNN:	k-Nearest neighbours
KR:	Kernel ridge
KRR:	Kernel ridge regression
kWp:	Kilowatt peak
LBT:	Ladybug tools
LCOE:	Levelised cost of electricity
LSTM:	Long short-term memory
MAE:	Mean absolute error
MAD:	Mean absolute deviation
MAPE:	Mean absolute percentage error
MASE:	Mean absolute scaled error
MedAE:	Median absolute error
ML:	Machine learning
MLP:	Multilayer perceptron
MPE:	Mean percentage error
MPFE:	Mean percent forecast error
MRE:	Mean relative error
MSLE:	Mean squared logarithmic error
MSE:	Mean squared error
MWp:	Megawatt peak
nMAE:	Normalised mean absolute error
nRMSE:	Normalised root mean square error
NAR:	Non-linear autoregressive
NARX:	Non-linear autoregressive exogenous
NN:	Neural network
NWP:	Numerical weather prediction
OLM:	Orientation-level modelling
PDPP:	Partial daily pattern prediction
PIML:	Physics-informed machine learning
PPE:	Potential, power and energy
PRISMA:	Preferred reporting items for systematic reviews and meta-analyses
PV:	Photovoltaic
PVPNet:	PV power prediction network
R^2:	Coefficient of determination
Rec-LSTM:	Recursive long short-term memory
RF:	Random forest
RMSD:	Relative mean square deviation
RMSE:	Root mean square error
RNN:	Recurrent neural network
RRMSE:	Relative root mean square error
rRMSE:	Relative root mean square error
RR:	Ridge regression
RSM:	Rooftop segmentation model
SBFM/SBFMs:	Similarity-based forecasting models
SAM:	Segment anything model
SDA:	Similar day analysis
SE:	Sample entropy
SLR:	Systematic literature review
sMAPE:	Symmetric mean absolute percentage error
SSA:	Singular spectrum analysis
SVM:	Support vector machine
TL:	Transfer learning
UNet/Residual UNet:	U-Net convolutional architecture
VGI:	Vertical global insolation
WAPE:	Weighted absolute percentage error
WMAE:	Weighted mean absolute error
XGBoost:	Extreme gradient boosting.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

Funding

The authors acknowledge the funding provided by Edinburgh Napier University in support of the PhD studies of Mohammed Sadeq.

References

[1] A. Faes, A. Virtuani, H. Quest, et al., "Building-Integrated Photovoltaics," *Nature Reviews Clean Technology* 1, no. 5 (2025): 333–350.

[2] H. Zhao, C. Liu, R. J. Yang, and C. Sun, "Large-Scale Prediction of Solar Irradiation, Shading Impacts, and Energy Generation on Building Façade Through Urban Morphological Indicators: A Machine Learning Approach," *Energy and Buildings* 323 (2024): 114797.

[3] B. D. Dimd, S. Völler, O.-M. Midtgård, U. Cali, and A. Sevault, "Quantification of the Impact of Azimuth and Tilt Angle on the Performance of a PV Output Power Forecasting Model for BIPVs," *IEEE Journal of Photovoltaics* 14, no. 1 (2024): 194–200.

[4] F. Wang, Z. Xuan, Z. Zhen, K. Li, T. Wang, and M. Shi, "A Day-Ahead PV Power Forecasting Method Based on LSTM-RNN Model and Time Correlation Modification Under Partial Daily Pattern Prediction Framework," *Energy Conversion and Management* 212 (2020): 112766.

[5] J. Polo, N. Martín-Chivelet, and C. Sanz-Saiz, "BIPV Modeling With Artificial Neural Networks: Towards a BIPV Digital Twin," *Energies* 15, no. 11 (2022): 4173.

[6] Y. Yue, Z. Yan, P. Ni, F. Lei, and G. Qin, "Promoting Solar Energy Utilization: Prediction, Analysis and Evaluation of Solar Radiation on Building Surfaces at City Scale," *Energy and Buildings* 319 (2024): 114561.

[7] A. Kumar, A. K. Dubey, I. S. Ramírez, A. M. del Río, and F. P. G. Márquez, "Artificial Intelligence Techniques for the Photovoltaic System: A Systematic Review and Analysis for Evaluation and Benchmarking," *Archives of Computational Methods in Engineering* 31, no. 8 (2024): 4429–4453.

ijer, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/er/1140262, Wiley Online Library on [20/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

[8] K. A. Tahir, "A Systematic Review and Evolutionary Analysis of the Optimization Techniques and Software Tools in Hybrid Microgrid Systems," *Energies* 18, no. 7 (2025): 1770.
[9] T. Tamer, I. G. Dino, and C. M. Akgül, "Data-Driven, Long-Term Prediction of Building Performance Under Climate Change: Building Energy Demand and BIPV Energy Generation Analysis Across Turkey," *Renewable and Sustainable Energy Reviews* 162 (2022): 112396.
[10] M. Diagne, M. David, P. Lauret, J. Boland, and N. Schmutz, "Review of Solar Irradiance Forecasting Methods and a Proposition for Small-Scale Insular Grids," *Renewable and Sustainable Energy Reviews* 27 (2013): 65–76.
[11] C. Voyant, G. Notton, S. Kalogirou, et al., "Machine Learning Methods for Solar Radiation Forecasting: A Review," *Renewable Energy* 105 (2017): 569–582.
[12] Y. Kashyap, A. Bansal, and A. K. Sao, "Solar Radiation Forecasting With Multiple Parameters Neural Networks," *Renewable and Sustainable Energy Reviews* 49 (2015): 825–835.
[13] J. Antonanzas, N. Osorio, R. Escobar, R. Urraca, F. J. Martinez-de-Pison, and F. Antonanzas-Torres, "Review of Photovoltaic Power Forecasting," *Solar Energy* 136 (2016): 78–111.
[14] F. Barbieri, S. Rajakaruna, and A. Ghosh, "Very Short-Term Photovoltaic Power Forecasting With Cloud Modeling: A Review," *Renewable and Sustainable Energy Reviews* 75 (2017): 242–263.
[15] U. K. Das, K. S. Tey, M. Seyedmahmoudian, et al., "Forecasting of Photovoltaic Power Generation and Model Optimization: A Review," *Renewable and Sustainable Energy Reviews* 81 (2018): 912–928.
[16] A. M. Assaf, H. Haron, H. N. A. Hamed, F. A. Ghaleb, S. N. Qasem, and A. M. Albarrak, "A Review on Neural Network Based Models for Short Term Solar Irradiance Forecasting," *Applied Sciences* 13, no. 14 (2023): 8332.
[17] R. Ahmed, V. Sreeram, Y. Mishra, and M. D. Arif, "A Review and Evaluation of the State-of-the-Art in PV Solar Power Forecasting: Techniques and Optimization," *Renewable and Sustainable Energy Reviews* 124 (2020): 109792.
[18] M. Massaoudi, I. Chihi, H. Abu-Rub, S. S. Refaat, and F. S. Oueslati, "Convergence of Photovoltaic Power Forecasting and Deep Learning: State-of-Art Review," *IEEE Access* 9 (2021): 136593–136615.
[19] M. N. Akhter, S. Mekhilef, H. Mokhlis, and N. M. Shah, "Review on Forecasting of Photovoltaic Power Generation Based on Machine Learning and Metaheuristic Techniques," *IET Renewable Power Generation* 13, no. 7 (2019): 1009–1023.
[20] M. Q. Raza, M. Nadarajah, and C. Ekanayake, "On Recent Advances in PV Output Power Forecast," *Solar Energy* 136 (2016): 125–144.
[21] A. Mellit, A. M. Pavan, E. Ogliari, S. Leva, and V. Lughi, "Advanced Methods for Photovoltaic Output Power Forecasting: A Review," *Applied Sciences* 10, no. 2 (2020): 487.
[22] J. R. Dos Reis, J. M. Tabora, M. C. de Lima, et al., "Medium and Long Term Energy Forecasting Methods: A Literature Review," *IEEE Access* 13 (2025): 29305–29326.
[23] Y. Xiao and M. Watson, "Guidance on Conducting a Systematic Literature Review," *Journal of Planning Education and Research* 39, no. 1 (2017): 93–112.
[24] H. A. M. Shaffril, A. A. Samah, S. F. Samsuddin, and Z. Ali, "Mirror-Mirror on the Wall, What Climate Change Adaptation Strategies Are Practiced by the Asian's Fishermen of All?" *Journal of Cleaner Production* 232 (2019): 104–117.
[25] D. Moher, A. Liberati, J. Tetzlaff, D. G. Altman, and for the PRISMA Group, "Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement," *BMJ* 339 (2009): b2535.
[26] W. Ali, W. U. Khan, M. A. Z. Raja, Y. He, and Y. Li, "Design of Nonlinear Autoregressive Exogenous Model Based Intelligence Computing for Efficient State Estimation of Underwater Passive Target," *Entropy* 23, no. 5 (2021): 550.
[27] A. Sedai, R. Dhakal, S. Gautam, et al., "Performance Analysis of Statistical, Machine Learning and Deep Learning Models in Long-Term Forecasting of Solar Power Production," *Forecasting* 5, no. 1 (2023): 256–284.
[28] S. Cantillo-Luna, R. Moreno-Chuquen, D. Celeita, and G. Anders, "Deep and Machine Learning Models to Forecast Photovoltaic Power Generation," *Energies* 16, no. 10 (2023): 4097.
[29] H. Lan, Z. Gou, and C. Hou, "Understanding the Relationship Between Urban Morphology and Solar Potential in Mixed-Use Neighborhoods Using Machine Learning Algorithms," *Sustainable Cities and Society* 87 (2022): 104225.
[30] G. Li, Z. Wang, C. Xu, et al., "A District-Scale Spatial Distribution Evaluation Method of Rooftop Solar Energy Potential Based on Deep Learning," *Solar Energy* 268 (2024): 112282.
[31] J. Tian and S. Xu, "A Morphology-Based Evaluation on Block-Scale Solar Potential for Residential Area in Central China," *Solar Energy* 221 (2021): 332–347.
[32] S. Xu, H. Jiang, F. Xiong, C. Zhang, M. Xie, and Z. Li, "Evaluation for Block-Scale Solar Energy Potential of Industrial Block and Optimization of Application Strategies: A Case Study of Wuhan, China," *Sustainable Cities and Society* 72 (2021): 103000.
[33] K. Zhao and Z. Gou, "Influence of Urban Morphology on Facade Solar Potential in Mixed-Use Neighborhoods: Block Prototypes and Design Benchmark," *Energy and Buildings* 297 (2023): 113446.
[34] K. H. Poon, J. H. Kämpf, S. E. R. Tay, N. H. Wong, and T. G. Reindl, "Parametric Study of URBAN Morphology on Building Solar Energy Potential in Singapore Context," *Urban Climate* 33 (2020): 100624.
[35] J. E. Fuentes, F. D. Moya, and O. D. Montoya, "Method for Estimating Solar Energy Potential Based on Photogrammetry From Unmanned Aerial Vehicles," *Electronics* 9, no. 12 (2020): 2144.
[36] H. Ren, C. Xu, Z. Ma, and Y. Sun, "A Novel 3D-Geographic Information System and Deep Learning Integrated Approach for High-Accuracy Building Rooftop Solar Energy Potential Characterization of High-Density Cities," *Applied Energy* 306 (2022): 117985.
[37] R. Zhu, C. Cheng, P. Santi, et al., "Optimization of Photovoltaic Provision in a Three-Dimensional City Using Real-Time Electricity Demand," *Applied Energy* 316 (2022): 119042.
[38] J. Tian, R. Ooka, and D. Lee, "Multi-Scale Solar Radiation and Photovoltaic Power Forecasting With Machine Learning Algorithms in Urban Environment: A State-of-the-Art Review," *Journal of Cleaner Production* 426 (2023): 139040.
[39] P. Wang, P. Yu, L. Huang, and Y. Zhang, "An Integrated Technical, Economic, and Environmental Framework for Evaluating the Rooftop Photovoltaic Potential of Old Residential Buildings," *Journal of Environmental Management* 317 (2022): 115296.

ijer, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/er/1140262, Wiley Online Library on [20/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

[40] T. Zhong, Z. Zhang, M. Chen, et al., "A City-Scale Estimation of Rooftop Solar Photovoltaic Potential Based on Deep Learning," *Applied Energy* 298 (2021): 117132.
[41] Q. Li, S. Krapf, Y. Shi, and X. X. Zhu, "SolarNet: A Convolutional Neural Network-Based Framework for Rooftop Solar Potential Estimation From Aerial Imagery," *International Journal of Applied Earth Observation and Geoinformation* 116 (2023): 103098.
[42] A. A. Tehrani, O. Veisi, B. V. Fakhr, and D. Du, "Predicting Solar Radiation in the Urban Area: A Data-Driven Analysis for Sustainable City Planning Using Artificial Neural Networking," *Sustainable Cities and Society* 100 (2024): 105042.
[43] Y. Zhang, A. Schlueter, and C. Waibel, "SolarGAN: Synthetic Annual Solar Irradiance Time Series on Urban Building Facades via Deep Generative Networks," *Energy and AI* 12 (2023): 100223.
[44] L. Yan, R. Zhu, M.-P. Kwan, et al., "Estimation of Urban-Scale Photovoltaic Potential: A Deep Learning-Based Approach for Constructing Three-Dimensional Building Models From Optical Remote Sensing Imagery," *Sustainable Cities and Society* 93 (2023): 104515.
[45] M. Díez-Mediavilla, M. C. Rodríguez-Amigo, M. I. Dieste-Velasco, T. García-Calderón, and C. Alonso-Tristán, "The PV Potential of Vertical Façades: A Classic Approach Using Experimental Data From Burgos, Spain," *Solar Energy* 177 (2019): 192–199.
[46] C. G. Ozoegwu, "Artificial Neural Network Forecast of Monthly Mean Daily Global Solar Radiation of Selected Locations Based on Time Series and Month Number," *Journal of Cleaner Production* 216 (2019): 1–13.
[47] C. Pan and J. Tan, "Day-Ahead Hourly Forecasting of Solar Generation Based on Cluster Analysis and Ensemble Model," *IEEE Access* 7 (2019): 112921–112930.
[48] J. Polo, N. Martín-Chivelet, M. Alonso-Abella, C. Sanz-Saiz, J. Cuenca, and M. de la Cruz, "Exploring the PV Power Forecasting at Building Façades Using Gradient Boosting Methods," *Energies* 16, no. 3 (2023): 1495.
[49] J. Jeong, D. Lee, and Y. T. Chae, "A Novel Approach for Day-Ahead Hourly Building-Integrated Photovoltaic Power Prediction by Using Feature Engineering and Simple Weather Forecasting Service," *Energies* 16, no. 22 (2023): 7477.
[50] N. Maitanova, J.-S. Telle, B. Hanke, et al., "A Machine Learning Approach to Low-Cost Photovoltaic Power Prediction Based on Publicly Available Weather Reports," *Energies* 13, no. 3 (2020): 735.
[51] W.-G. Shin, J.-Y. Shin, H.-M. Hwang, C.-H. Park, and S.-W. Ko, "Power Generation Prediction of Building-Integrated Photovoltaic System With Colored Modules Using Machine Learning," *Energies* 15, no. 7 (2022): 2589.
[52] D. Markovics and M. J. Mayer, "Comparison of Machine Learning Methods for Photovoltaic Power Forecasting Based on Numerical Weather Prediction," *Renewable and Sustainable Energy Reviews* 161 (2022): 112364.
[53] L. Visser, T. AlSkaif, and W. van Sark, "Operational Day-Ahead Solar Power Forecasting for Aggregated PV Systems With a Varying Spatial Distribution," *Renewable Energy* 183 (2022): 267–282.
[54] S. A. Eroshenko, A. I. Khalyasmaa, D. A. Snegirev, V. V. Dubailova, A. M. Romanov, and D. N. Butusov, "The Impact of Data Filtration on the Accuracy of Multiple Time-Domain Forecasting for Photovoltaic Power Plants Generation," *Applied Sciences* 10, no. 22 (2020): 8265.
[55] H. Sangrody, N. Zhou, and Z. Zhang, "Similarity-Based Models for Day-Ahead Solar PV Generation Forecasting," *IEEE Access* 8 (2020): 104469–104478.
[56] W. Zhao, H. Zhang, J. Zheng, et al., "A Point Prediction Method Based Automatic Machine Learning for Day-Ahead Power Output of Multi-Region Photovoltaic Plants," *Energy* 223 (2021): 120026.
[57] S. S. kumar, A. Karthick, R. Shankar, and G. Dharmaraj, "Energy Forecasting of the Building Integrated Photovoltaic System Based on Deep Learning Dragonfly-Firefly Algorithm," *Energy* 308 (2024): 132926.
[58] R. Kabilan, V. Chandran, J. Yogapriya, et al., "Short-Term Power Prediction of Building Integrated Photovoltaic (BIPV) System Based on Machine Learning Algorithms," *International Journal of Photoenergy* 2021 (2021): 5582418.
[59] S. Kaliappan, R. Saravanakumar, A. Karthick, et al., "Hourly and Day Ahead Power Prediction of Building Integrated Semitransparent Photovoltaic System," *International Journal of Photoenergy* 2021 (2021): 7894849.
[60] Z. A. Khan, T. Hussain, and S. W. Baik, "Dual Stream Network With Attention Mechanism for Photovoltaic Power Forecasting," *Applied Energy* 338 (2023): 120916.
[61] A. Bellagarda, D. Grassi, A. Aliberti, L. Bottaccioli, A. Macii, and E. Patti, "Effectiveness of Neural Networks and Transfer Learning to Forecast Photovoltaic Power Production," *Applied Soft Computing* 149 (2023): 110988.
[62] Q. Li, Y. Xu, B. S. H. Chew, H. Ding, and G. Zhao, "An Integrated Missing-Data Tolerant Model for Probabilistic PV Power Generation Forecasting," *IEEE Transactions on Power Systems* 37, no. 6 (2022): 4447–4459.
[63] T. Polasek and M. Čadík, "Predicting Photovoltaic Power Production Using High-Uncertainty Weather Forecasts," *Applied Energy* 339 (2023): 120989.
[64] L. Cheng, H. Zang, T. Ding, Z. Wei, and G. Sun, "Multi-Meteorological-Factor-Based Graph Modeling for Photovoltaic Power Forecasting," *IEEE Transactions on Sustainable Energy* 12, no. 3 (2021): 1593–1603.
[65] Y. Qu, J. Xu, Y. Sun, and D. Liu, "A Temporal Distributed Hybrid Deep Learning Model for Day-Ahead Distributed PV Power Forecasting," *Applied Energy* 304 (2021): 117704.
[66] M. AlShafeey and C. Csáki, "Evaluating Neural Network and Linear Regression Photovoltaic Power Forecasting Models Based on Different Input Methods," *Energy Reports* 7 (2021): 7601–7614.
[67] H. Zang, L. Cheng, T. Ding, K. W. Cheung, Z. Wei, and G. Sun, "Day-Ahead Photovoltaic Power Forecasting Approach Based on Deep Convolutional Neural Networks and Meta Learning," *International Journal of Electrical Power & Energy Systems* 118 (2020): 105790.
[68] D. Lee, J. Jeong, S. H. Yoon, and Y. T. Chae, "Improvement of Short-Term BIPV Power Predictions Using Feature Engineering and a Recurrent Neural Network," *Energies* 12, no. 17 (2019): 3247.
[69] C.-J. Huang and P.-H. Kuo, "Multiple-Input Deep Convolutional Neural Network Model for Short-Term Photovoltaic Power Forecasting," *IEEE Access* 7 (2019): 74822–74834.
[70] L. Wen, K. Zhou, S. Yang, and X. Lu, "Optimal Load Dispatch of Community Microgrid With Deep Learning Based Solar Power and Load Forecasting," *Energy* 171 (2019): 1053–1065.
[71] A. Khandakar, M. E. H. Chowdhury, M.-K. Kazi, et al., "Machine Learning Based Photovoltaics (PV) Power Prediction Using Different Environmental Parameters of Qatar," *Energies* 12, no. 14 (2019): 2782.

ijer, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/er/1140262, Wiley Online Library on [20/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

[72] S. Sarkar, A. Karthick, V. K. Chinnaiyan, and P. P. Patil, "Energy Forecasting of the Building-Integrated Photovoltaic Facade Using Hybrid LSTM," *Environmental Science and Pollution Research* 30, no. 16 (2023): 45977–45985.

[73] M. Ragupathi and R. Ramasubbu, "Performance Prediction of Building Integrated Photovoltaic System Using Hybrid Deep Learning Algorithm," *International Journal of Photoenergy* 2022 (2022): 6111030.

[74] L. Zhang, J. Wang, Y. Qian, and Z. Li, "Photovoltaic Power Prediction System Based on Multi-Stage Data Processing Strategy and Improved Optimizer," *Applied Mathematical Modelling* 132 (2024): 226–251.

[75] D. Rangelov, M. Boerger, N. Tcholtchev, P. Lämmel, and M. Hauswirth, "Design and Development of a Short-Term Photovoltaic Power Output Forecasting Method Based on Random Forest, Deep Neural Network and LSTM Using Readily Available Weather Features," *IEEE Access* 11 (2023): 41578–41595.

[76] M. A. Houran, S. M. S. Bukhari, M. H. Zafar, M. Mansoor, and W. Chen, "COA-CNN-LSTM: Coati Optimization Algorithm-Based Hybrid Deep Learning Model for PV/Wind Power Forecasting in Smart Grid Applications," *Applied Energy* 349 (2023): 121638.

[77] B. Chen, P. Lin, Y. Lai, S. Cheng, Z. Chen, and L. Wu, "Very-Short-Term Power Prediction for PV Power Plants Using a Simple and Effective RCC-LSTM Model Based on Short Term Multivariate Historical Datasets," *Electronics* 9, no. 2 (2020): 289.

[78] Y. Zhou, N. Zhou, L. Gong, and M. Jiang, "Prediction of Photovoltaic Power Output Based on Similar Day Analysis, Genetic Algorithm and Extreme Learning Machine," *Energy* 204 (2020): 117894.

[79] C. Cui, Y. Zou, L. Wei, and Y. Wang, "Evaluating Combination Models of Solar Irradiance on Inclined Surfaces and Forecasting Photovoltaic Power Generation," *IET Smart Grid* 2, no. 1 (2019): 123–130.

[80] L.-L. Li, S.-Y. Wen, M.-L. Tseng, and C.-S. Wang, "Renewable Energy Prediction: A Novel Short-Term Prediction Model of Photovoltaic Output Power," *Journal of Cleaner Production* 228 (2019): 359–375.

[81] A. Nespoli, E. Ogliari, S. Leva, et al., "Day-Ahead Photovoltaic Forecasting: A Comparison of the Most Effective Techniques," *Energies* 12, no. 9 (2019): 1621.

[82] L. Liu, M. Zhan, and Y. Bai, "A Recursive Ensemble Model for Forecasting the Power Output of Photovoltaic Systems," *Solar Energy* 189 (2019): 291–298.

[83] L. Fara, A. Diaconu, D. Craciunescu, and S. Fara, "Forecasting of Energy Production for Photovoltaic Systems Based on ARIMA and ANN Advanced Models," *International Journal of Photoenergy* 2021 (2021): 6777488.

[84] E. Sarmas, S. Strompolas, V. Marinakis, F. Santori, M. A. Bucarelli, and H. Doukas, "An Incremental Learning Framework for Photovoltaic Production and Load Forecasting in Energy Microgrids," *Electronics* 11, no. 23 (2022): 3962.

[85] I. Abouelaziz and Y. Jouane, "Photogrammetry and Deep Learning for Energy Production Prediction and Building-Integrated Photovoltaics Decarbonization," *Building Simulation* 17, no. 2 (2024): 189–205.

[86] M. Jasinski, O. Homaee, D. Opałkowski, A. Najafi, and Z. Leonowicz, "On the Forecastability of Solar Energy Generation by Rooftop Panels Pointed in Different Directions," *IEEE Transactions on Sustainable Energy* 15, no. 1 (2024): 699–702.

[87] H. Ni, D. Wang, W. Zhao, et al., "Enhancing Rooftop Solar Energy Potential Evaluation in High-Density Cities: A Deep Learning and GIS Based Approach," *Energy and Buildings* 309 (2024): 113743.

[88] Z. Zhen, J. Liu, Z. Zhang, et al., "Deep Learning Based Surface Irradiance Mapping Model for Solar PV Power Forecasting Using Sky Image," *IEEE Transactions on Industry Applications* 56, no. 4 (2020): 3385–3396.

[89] Y. Zhou, S. Zheng, and G. Zhang, "Machine Learning-Based Optimal Design of a Phase Change Material Integrated Renewable System With on-Site PV, Radiative Cooling and Hybrid Ventilations—Study of Modelling and Application in Five Climatic Regions," *Energy* 192 (2020): 116608.

[90] Y. Zhou, S. Zheng, and G. Zhang, "Machine-Learning Based Study on the on-Site Renewable Electrical Performance of an Optimal Hybrid PCMs Integrated Renewable System With High-Level Parameters' Uncertainties," *Renewable Energy* 151 (2020): 403–418.

[91] S. O. Abter, S. M. Jameel, H. M. Majeed, and A. H. Sabry, "Intelligent Forecasting Temperature Measurements of Solar PV Cells Using Modified Recurrent Neural Network," *EUREKA: Physics and Engineering* 3 (2024): 169–177.

[92] K. Dong, Q. Yu, Z. Guo, et al., "Advancing Building Facade Solar Potential Assessment Through AIoT, GIS, and Meteorology Synergy," *Advances in Applied Energy* 17 (2025): 100212.

[93] M. Sadeq and S. Abdellatif, "PV-ON: An Online/Bilingual PV Sizing Tool for Grid-Connected System, Case Studies in Egypt," *International Transactions on Electrical Energy Systems* 31, no. 7 (2021): e12910.

[94] M. Fariz, M. Sadeq, and N. Sellami, "Integrated Simulation and Optimisation Framework for Techno-Economic Evaluation of BIPV Systems: A Scottish Case Study," *e-Prime - Advances in Electrical Engineering, Electronics and Energy* 13 (2025): 101091.

[95] H. I. Hazim, K. A. Baharin, C. K. Gan, and A. H. Sabry, "Techno-Economic Optimization of Photovoltaic (PV)-Inverter Power Sizing Ratio for Grid-Connected PV Systems," *Results in Engineering* 23 (2024): 102580.

[96] Z. R. K. Abojela, M. K. M. Desa, and A. H. Sabry, "Current Prospects of Building-Integrated Solar PV Systems and the Application of Bifacial PVs," *Frontiers in Energy Research* 11 (2023): 1164494.

[97] Y. Zhou, "Demand Response Flexibility With Synergies on Passive PCM Walls, BIPVs, and Active Air-Conditioning System in a Subtropical Climate," *Renewable Energy* 199 (2022): 204–225.

[98] S. Bak, S. Choi, D. Yang, D. Kim, H. Rho, and K. Lee, "Transfer Learning for Photovoltaic Power Forecasting Across Regions Using Large-Scale Datasets," *IEEE Access* 13 (2025): 136175–136190.

[99] S. Tajjour, S. S. Chandel, H. Malik, F. P. G. Márquez, and M. A. Alotaibi, "Daily Power Generation Forecasting for a Grid-Connected Solar Power Plant Using Transfer Learning Technique," *Applied Intelligence* 55, no. 6 (2025): 383.

[100] J. Li, H. Xie, J. Zhang, et al., "A Systematic Review of Reinforcement Learning in Building-Integrated Photovoltaic (BIPV) Optimization," *Advances in Applied Energy* 19 (2025): 100239.

[101] M. F. Hanif, R. Ahmad, A. B. Farooq, et al., "The Solar AI Nexus: Reinforcement Learning Shaping the Future of Energy Management," *WIREs Energy and Environment* 14, no. 3 (2025): e70012.

ijer, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/er/1140262, Wiley Online Library on [20/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

[102] Y. Yang, Y. Liu, Y. Zhang, S. Shu, and J. Zheng, "DEST-GNN: A Double-Explored Spatio-Temporal Graph Neural Network for Multi-Site Intra-Hour PV Power Forecasting," *Applied Energy* 378 (2025): 124744.
[103] J. Yu, X. Li, L. Yang, et al., "Deep Learning Models for PV Power Forecasting: Review," *Energies* 17, no. 16 (2024): 3973.
[104] S. M. Parsa, "Physics-Informed Machine Learning Meets Renewable Energy Systems: A Review of Advances, Challenges, Guidelines, and Future Outlooks," *Applied Energy* 402 (2025): 126925.
[105] K. Wang, L. Wang, Q. Meng, et al., "Accurate Photovoltaic Power Prediction via Temperature Correction With Physics-Informed Neural Networks," *Energy* 328 (2025): 136546.
[106] M. Wang and C. Xiang, "Machine Learning for Power Forecasting in BIPV and BAPV Systems: A Review," *Journal of Building Engineering* 113 (2025): 114117.
[107] A. Duran, C. Waibel, V. Piccioni, B. Bickel, and A. Schlueter, "A Review on Artificial Intelligence Applications for Facades," *Building and Environment* 269 (2025): 112310.

ijer, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/er/1140262, Wiley Online Library on [20/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License